

## Subject Description Form

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| <b>Subject Code</b>                                   | ME41006                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Subject Title</b>                                  | Perceptual Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Credit Value</b>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Level</b>                                          | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Pre-requisite/<br/>Co-requisite/<br/>Exclusion</b> | Pre-requisite: ME31002 Linear Systems and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Objectives</b>                                     | <p>The subject aims to equip students with knowledge of:</p> <ol style="list-style-type: none"> <li>1. Artificial robot perception</li> <li>2. Perception-guided control</li> <li>3. Adaptive robot behaviour</li> <li>4. Perception-aided algorithms</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Intended Learning Outcomes</b>                     | <p>Upon completion of the subject, students will be able to:</p> <ol style="list-style-type: none"> <li>a) Able to mathematically model the different perceptual modalities used for robotic systems;</li> <li>b) Able to design perception-guided motion controls for mechanical robots</li> <li>c) Able to use perceptual feedback to implement adaptive robot behaviours</li> <li>d) Able to design perception-aided methods for learning properties about the environment</li> <li>e) Able to conduct experiments with perceptual and robotic systems</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b>      | <p><b>Artificial robot perception.</b> Vision sensors (monocular perception and RGB-D sensors), thermal imaging (models and principles), touch (force and tactile imaging), proximity (different ranging methodologies), audio sensing.</p> <p><b>Perception-guided control.</b> Sensor-motion coordination problem, derivation of sensorimotor models (analytical and computational), formulation of sensor servoing controls (vision-based, thermal-based, touch-based, proximity-based).</p> <p><b>Adaptive robot behaviour.</b> Braitenberg machines, reactive motion paradigms (potential fields, subsumption architecture, etc.), hybrid paradigms, multi-agent systems, robot babbling, bug algorithms, sensor-based navigation.</p> <p><b>Perception-aided algorithms.</b> Iterative closest point (ICP), simultaneous localisation and mapping (SLAM), sensor-based model learning, and image registration.</p> <p><b>Practical work.</b> Various laboratory sessions will be delivered to provide students with hands-on experience.</p> |

| <b>Teaching/Learning Methodology</b>                                     | <div><div><div>1. Lectures aim at providing students with fundamental knowledge required for understanding and analysing different perceptual robotic systems, including its mathematical models, controller design, and algorithms. (Outcomes a to d)</div><div>2. Tutorials aim at enhancing students’ analytical and problem solving skills on robotics. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes a to d)</div><div>3. The experiments/project aim to provide hands-on experience for developing perceptual robots, and reinforcing the acquired knowledge. (Outcomes a to e)</div></div><table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Intended Subject Learning Outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>2. Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>3. Experiments/Project</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table></div>                                                                                                                                                                                                                                                                               | Teaching/Learning Methodology     | Intended Subject Learning Outcomes to be assessed |                                                                                |         |                          |        | a                           | b | c | d | e | 1. Lecture | √             | √   | √ | √ |   | 2. Tutorial | √ | √                                                                        | √   | √ |   | 3. Experiments/Project | √ | √ | √                    | √   | √ |   |   |   |  |       |       |  |  |  |  |  |
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| Teaching/Learning Methodology                                            | Intended Subject Learning Outcomes to be assessed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                   |                                                   |                                                                                |         |                          |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
|                                                                          | a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | b                                 | c                                                 | d                                                                              | e       |                          |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| 1. Lecture                                                               | √                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | √                                 | √                                                 | √                                                                              |         |                          |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| 2. Tutorial                                                              | √                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | √                                 | √                                                 | √                                                                              |         |                          |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| 3. Experiments/Project                                                   | √                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | √                                 | √                                                 | √                                                                              | √       |                          |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| <b>Assessment Methods in Alignment with Intended Learning Outcomes</b>   | <div><table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Class test</td><td>10%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Coursework: May include assignments, project, experiments, lab report</td><td>50%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Final Examination</td><td>40%</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr></table><p>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:</p><p>Overall Assessment:</p><p>0.40 x End of Subject Examination + 0.60 x Continuous Assessment.</p><div><div>1. The continuous assessment aims at evaluating the progress of the students’ study, assisting them in self-monitoring the respective learning outcomes, and applying the knowledge learnt in practical situations.</div><div>2. The examination is used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes.</div></div></div> | Specific assessment methods/tasks | % weighting                                       | Intended subject learning outcomes to be assessed (Please tick as appropriate) |         |                          |        |                             | a | b | c | d | e          | 1. Class test | 10% | √ | √ | √ | √           | √ | 2. Coursework: May include assignments, project, experiments, lab report | 50% | √ | √ | √                      | √ | √ | 3. Final Examination | 40% | √ | √ | √ | √ |  | Total | 100 % |  |  |  |  |  |
| Specific assessment methods/tasks                                        | % weighting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                   |                                                   | Intended subject learning outcomes to be assessed (Please tick as appropriate) |         |                          |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
|                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | a                                 | b                                                 | c                                                                              | d       | e                        |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| 1. Class test                                                            | 10%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | √                                 | √                                                 | √                                                                              | √       | √                        |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| 2. Coursework: May include assignments, project, experiments, lab report | 50%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | √                                 | √                                                 | √                                                                              | √       | √                        |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| 3. Final Examination                                                     | 40%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | √                                 | √                                                 | √                                                                              | √       |                          |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| Total                                                                    | 100 %                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                   |                                                   |                                                                                |         |                          |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| <b>Student Study Effort Expected</b>                                     | <table><tr><td>Class contact:</td><td></td></tr><tr><td>▪ Lectures</td><td>33 Hrs.</td></tr><tr><td>▪ Tutorials / Laboratory</td><td>6 Hrs.</td></tr><tr><td>Other student study effort:</td><td></td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Class contact:                    |                                                   | ▪ Lectures                                                                     | 33 Hrs. | ▪ Tutorials / Laboratory | 6 Hrs. | Other student study effort: |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| Class contact:                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                   |                                                   |                                                                                |         |                          |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| ▪ Lectures                                                               | 33 Hrs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                   |                                                   |                                                                                |         |                          |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| ▪ Tutorials / Laboratory                                                 | 6 Hrs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                   |                                                   |                                                                                |         |                          |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |
| Other student study effort:                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                   |                                                   |                                                                                |         |                          |        |                             |   |   |   |   |            |               |     |   |   |   |             |   |                                                                          |     |   |   |                        |   |   |                      |     |   |   |   |   |  |       |       |  |  |  |  |  |

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|                                    | ▪ Course work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 40 Hrs.  |
|                                    | ▪ Self-learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 36 Hrs.  |
|                                    | Total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 115 Hrs. |
| <b>Reading List and References</b> | 1. Lecture Notes, Articles, and Handouts<br>2. Computer Vision: A Modern Approach, David A. Forsyth and Jean Ponce, latest edition.<br>3. Introduction to AI Robotics, Robin Murphy, MIT Press Cambridge, MA, USA, latest edition.<br>4. Principles of Robot Motion: Theory, Algorithms, and Implementations, Howie Choset et al, MIT, latest edition.<br>5. Vehicles: Experiments in Synthetic Psychology, Valentino Braitenberg, MIT Press Ltd, latest edition.<br>6. Robotics Modelling, Planning and Control, Bruno Siciliano et al, latest edition. |          |
| <b>Last Update</b>                 | July 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |